Pauline Emma Michèle De Baets

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Profile

EPFL Microengineering student and Hardware Engineering Intern at RIVR (an Amazon company), currently focused on designing a fully automated 24/7 lifecycle test rig and on end-of-line validation and debugging of legged-wheeled robots. Solid experience in embedded systems, mechanical design and hands-on hardware, including rocket recovery mechanisms.

Education

EPFL – BSc in Microengineering (2023–2026) GPA: 4.94/6
Relevant coursework at EPFL (/6): Mechanism Conception I & II (5.0, 6.0), Analysis I & II (5.5, 6.0), Linear Algebra (5.0), General Physics: Mechanics (5.0), Logic Systems (6.0), Microcontrollers (5.25).
Third-year exchange at TU Delft (Electrical Engineering Faculty) (2025–2026).
Relevant coursework at TU Delft (/10): Signals and Systems (9.0), Sensors and Actuators (MSc, 8.0), Systems and Control (7.5), Engineering Optimization (MSc, 7.5).
Collège Sismondi, Geneva – Bilingual Maturité (2019–2023), physics and applied mathematics GPA: 5.8/6

Experience

Hardware Engineering Intern – RIVR (an Amazon company), Testing Team (Aug 2026–present, 6 months)

- Designing a fully automated, 24/7 high-cycle lifecycle test rig for the robot's payload mechanism.
- Responsible for end-of-line (EOL) testing of robots before shipment to customers, including root-cause debugging of failures, mostly in hardware components, using Foxglove and manual analysis of ROS bag data.
- Developed a user-friendly Python-based EOL test interface that connects to the robot over SSH, gives direct acces to bring-up and homing, matches ROS bags to individual mechanical stress tests, automatically analyses the recorded data, generates test reports and uploads results to SharePoint for validation by the testing team; now also adopted by another international team for its EOL tests.
- Currently extending it into an automated EOL test sequence that drives the robot through ROS 2 commands.
- **Internship - Creolab Orthodontics Laboratory** (2022, 1 month): 3D scanning, model segmentation and high-precision resin printing.
- **Volunteer - Ilanga** (2022, 1.5 months): taught daily English classes to children and teenagers (ages 3–18) in a community programme supporting destitute children and mothers.

Technical Skills & Other

- **Programming languages:** C++, C, Python, AVR assembly.
- **Robotics & Testing:** ROS 2 (robot commands, ROS bag data analysis), Foxglove, SSH, hardware debugging (IMUs, actuators, lidars, Jetson boards), test rig design, automated data analysis and test reporting.
- **Embedded & Electronics:** Embedded systems (ATmega128L), microcontrollers, basic digital/analog electronics.
- **Mechanical & CAD:** 3D/2D CAD (SolidWorks, CATIA), mechanical design, 3D printing.
- **Other:** Introductory machine learning (radar-based gesture recognition, pedestrian detection; Python, NumPy).
- **Interests:** Football, tennis, alpine skiing, and piano.

Languages

French: Native • **English:** C2 (Cambridge Proficiency) • **German:** B2 • **Dutch:** Intermediate

Selected Engineering Projects

- **Line cutter (Rocket Team):** Designed, manufactured, iterated and tested a compact pyrotechnic device using an e-match–triggered charge and a guided blade to cut high-strength Dyneema reefing lines, triggering the second phase of the rocket's descent.
- **Force Sensor - Mechanism Conception Group Project (EPFL):** Design of a 3-DOF force sensor capable of measuring contact forces in all directions with high precision, using compliant mechanisms and flexures, force and moment balancing and capacitive displacement sensors. Developed the full mechanical concept, modelling and dimensioning to meet industrial constraints from Hexagon.
- **Pac-Man Style Game on ATmega128L (EPFL):** Built a full Pac-Man style game on the ATmega128L entirely in AVR assembly, integrating WS2812B LED matrix graphics, a PS/2 keyboard user interface, LCD display, and an internal EEPROM profile storage system.
- **Radar-based hand gesture recognition (TU Delft):** Developed a machine learning pipeline for automatic classification of four hand gestures (wave, pinch, swipe, click) from micro-Doppler radar spectrograms (DopNet dataset), including preprocessing, feature extraction, and a neural network evaluated on an unseen test set.

Relevant project grades: Force Sensor 6/6, Pac-Man 5.5/6, Radar-based hand gesture recognition 8/10

Extracurricular

- **EPFL Rocket Team - Recovery Division** (2024-2025): responsible for the design, manufacturing and testing of the line cutter system.
- **Robopoly - EPFL Robotics Association** (2023-2025): PCB soldering and assembly, electronics prototyping.